

We shall now look at two of the sectors of the economy in this unit. One is the agricultural sector that provides livelihood to roughly two-thirds of the total workforce in the country. The other is the industrial sector, which consists of manufacturing, power, gas and water supply.

22.2 AGRICULTURAL DATA

You are already aware of the importance of agriculture to the Indian economy and indeed to the Indian way of life. You would, therefore, like to examine several aspects of agriculture like the level of production of different crops and commodities, the availability and utilization of important inputs for agricultural production, incentives, availability of post-harvest services and the role of agriculture in development. Similarly, you would like to know about the livestock and their products, fisheries and forestry, people engaged in these activities and so on. All these analyses require enormous amount of data over time and space. Let us have a look at what kind of data are available and where.

The **Directorate of Economics and Statistics (DESMOA)** in the **Department of Agriculture and Cooperation** (website <http://eands.dacnet.nic.in>) and the **Animal Husbandry Statistics Division (AHSD)** of the **Department of Animal Husbandry, Dairy and Fisheries** (website <http://dahd.nic.in/dahd/statistics/animal-husbandry-statistics.aspx>) in the Ministry of Agriculture are the major sources of data on agriculture and allied activities. Among the major efforts at collection of data mounted at regular intervals, the following are the major ones:

- i) the quinquennial agricultural census and the input survey;
- ii) the cost of cultivation studies;
- iii) annual estimates of crop production;
- iv) the quinquennial livestock census; and
- v) integrated sample survey to estimate the production of major livestock products.

DESMOA releases statistics on agriculture and allied activities flowing from its work and also collected from other division (like the **Agricultural Census Division**) and Departments of the Ministry of Agriculture and other Ministries and agencies in their comprehensive annual publication **Agricultural Statistics at a Glance (ASG)**¹. Other publications include, “**Cost of Cultivation in India**”, the monthly bulletin “**Agricultural Situation In India**”. **AHSD** publishes the data flowing from its activities [which include the activities mentioned at (iv) and (v) above] and those collected from other divisions of the Department of Animal Husbandry, Dairy and Fisheries and other agencies in its biennial publication **Basic Animal Husbandry Statistics (BAHS)**.

22.2.1 Agricultural Census

Agriculture Census forms part of a broader system of collection of Agricultural Statistics. It is a large-scale statistical operation for the collection and derivation of quantitative information about the structure of agriculture in the country. An

¹ This can be accessed in the **Ministry of Agriculture (Department of Agriculture) website** . One can also access www.data.gov.in and click on

agricultural operational holding is the ultimate unit for taking decision for development of Agriculture at micro level. It is for this reason that an operational holding is taken as the statistical unit of data collection for describing the structure of agriculture. Through Agriculture Census it is endeavored to collect basic data on important aspects of agricultural economy for all the operational holdings in the country. Aggregation of data is done at various levels of administrative units. The agricultural census started in 1970-71. The ninth of these censuses relates to the reference year 2010-11.

Periodic Agriculture Censuses are important as these are the main source of information on basic characteristics of operational holdings such as land use and cropping patterns, irrigation status, tenancy particulars and the terms of leasing. This information is tabulated by different size classes and social groups including Scheduled Castes/Scheduled Tribes which are needed for development planning, socio-economic policy formulation and establishment of national priorities. The census also provides the basis for the development of a comprehensive integrated national system of agricultural statistics and has links with various components of the national statistical system. The whole project of Agriculture Census in the country is implemented in three distinct phases, which are statistically linked together but focus on different aspects of agricultural statistics. In Phase-I, a list of holdings with their area and social characteristics and gender of the holders is prepared. In Phase-II, detailed data on agricultural characteristics of holdings are collected from selected villages. Thus the whole operation of Agriculture Census in India is not really a complete Census. In fact, it is a combination of Census and Sample Survey.²

The third phase, which is better known as the **Input Survey** and conducted in the year following the census year³, relates to collection of data on the pattern of input-use across crops, regions and size-groups of holdings. The main objective of the input survey is to generate data on consumption of various agricultural inputs, according to major size-groups of operational holdings, viz., marginal (below 1 ha.), small (1- 1.99 ha.), semi-medium (2- 3.99 ha.), medium (4- 9.99 ha.) and large (10 ha. and above), for getting an insight into the consumption pattern of inputs by various categories of farmers. This information is vital for planning production, imports and distribution of fertilizers. The inputs covered in the survey include chemical fertilizers, HYV seeds, pesticides, farmyard manures/compost, bio-fertilizers, agricultural implements and machinery, livestock and agricultural credit besides data on input use including use of certified/notified seeds, high yielding variety seeds, pest control measurements adopted by cultivators, educational qualification, age and size of households of operational holders are captured through Input Survey. The results of the first phase, including the all-India report of the first phase of the ninth agricultural census can be accessed at <http://agcensus.nic.in/agcen201011.html>. The provisional results of the second phase have also been released. Data collection and processing of the third phase is progressing in the different States and UTs. Selective information is also available on the **internet**⁴ through a query-based facility and can provide data on operational

² This part is reproduced from <http://agcensus.dacnet.nic.in/>, which can be accessed to get the latest data, up to tehsil level from Agricultural Census 2013.

³ The input survey was initiated in the 1975-76 Agricultural Census and conducted in 1976-77.

⁴ On the website <http://agcensus.dacnet.nic.in/nationalholdingtype.aspx>, one can access tables related to agricultural census years 1995-96 to 2010-11. Further, On the website <http://inputsurvey.dacnet.nic.in/nationalsummary.aspx> one can access the tables related to 1996-97 to 2006-07 so far.

holdings at the national, State, district and *tehsil* levels by type, characteristics and size class/group. You can, for example, get information on operational holdings by size class of the holding, social group (Scheduled Caste, Scheduled Tribe, Other and all groups) and tenancy characteristics of the holding (lease terms, land use, irrigation status, irrigation by sources, wells and tube wells, gross cropped area, cropping pattern, and area dispersal. Clearly, there is a time lag in the availability of agricultural census and Input Survey results.

22.2.2 Studies on Cost of Cultivation

DESMOA implements a comprehensive scheme for studying the cost of cultivation of principal crops in India and this results in the collection and compilation of field data on the cost of cultivation and production in respect of 29 principal crops leading to estimates of crop-wise and State-wise costs of cultivation and also computation of the *index of the terms of trade between agriculture and non-agricultural sectors*. The scheme covers 19 States and foodgrain crops, oil seeds and commercial crops and selected vegetables. Statistics of cost of cultivation of principal crops are published by **DESMOA** in its publication “**Cost of Cultivation in India**” and also in **ASG**. The **Commission for Agricultural Costs and Prices (CACP)** makes use of the estimates of cost of cultivation and production and the structure of cost of cultivation flowing from these studies, along with an analysis of a wide spectrum of data on variables like market prices, productivity of the crops concerned, domestic and global inter-crop price parity, the terms of trade between the agricultural and the non-agricultural sectors and the supply-demand situation in arriving at their recommendations to Government on Minimum Support Prices (MSP). The **CACP reports** thus contain not only the data thrown up by the cost of cultivation studies but also those assembled by it from various sources for its work. For example any of the **CACP report** presents:

- i) estimates of cost of cultivation per hectare, cost of production per quintal, and the value of the main product and the byproduct;
- ii) the breakup of the cost by inputs like human labour, bullock labour, machine labour, seed, fertilizers and manures, insecticides, irrigation charges and so on;
- iii) a variable input price index; and
- iv) month-end wholesale prices of the commodities concerned by variety and location.

for all the commodities dealt with in the report and for all states. In addition, it gives information on the index number of whole sale prices for farm inputs, index numbers of whole sale prices of cereals, pulses, etc., and data on a host of aspects relevant to the subject matter of the report like area, production, yield of major crops, buffer stock of cereals and month-wise and area-wise daily wage rates for agricultural labour⁵. MSPs for different commodities are also published in **ASG**.

⁵ The cost of cultivation studies reports and related data can be accessed at http://eands.dacnet.nic.in/Cost_of_Cultivation.htm. Now, the data is available for reference years 2004-05 to 2011-12. The details regarding the work of CACP and the data presented in **CACP reports** are based on the information available in the **CACP website** <http://cacp.dacnet.nic.in>. The MSP up to agricultural year 2014-15 can be viewed in the link <http://cacp.dacnet.nic.in/ViewContents.aspx?Input=1&PageId=36&KeyId=0A>.

The other useful outcome of the cost of cultivation studies, as already mentioned, is the **index of the terms of trade between the agricultural and non-agricultural sectors**. This is defined as

$$\text{Index of the Terms of Trade} = \frac{\text{IPR [Index of Prices Received]}}{\text{IPP [Index of Prices Paid]}}$$

(Base Year : the triennium ending 1991-92)

The numerator relates to the prices that the agricultural sector gets for its produce while the denominator relates to the prices that the sector pays for products it purchases for final consumption, intermediate consumption and capital formation. The time series of ITT from 1981-82 onwards, complete with individual values of IPR and IPP and those of the component indices of IPP, namely, final consumption (weight in IPP 73.54), intermediate consumption (weight in IPP 21.63) and capital formation (weight in IPP 4.83) are published in **ASG**.

The Terms of Trade, specifically, the terms of trade between agriculture and industry is the ratio of agricultural prices to industrial prices, both measured as price indices. The terms of trade between agricultural and industry lies at the heart of the government's policy in the agricultural sector. A rise in the ratio (agricultural prices divided by industrial prices) means that the agricultural sector is better off in terms of its purchasing power of industrial goods. Suppose there are just two goods in the economy, bananas and pins and you produce and sell bananas. Both initially cost Rs. 10. If now both prices go up by 100 per cent, nothing happens to the terms of trade and purchasing power remains unchanged. If instead, banana prices go up by 100 per cent and the price of pins doesn't, then as a banana manufacturer you are better off since one banana can now buy two pins instead of one. The ratio or the terms of trade between bananas and pins has doubled. Substitute agriculture for bananas and industry for pins and you can get what a rise in agriculture's terms of trade does for farmers.⁶

The National Statistical Commission has pointed out that there is a sizeable time lag in the availability of results for use in the work of CACP and that steps should be taken to cut down delays in the release of the results of these studies⁷.

22.2.3 Annual Estimates of Crop Production

DESMOA makes and releases annual estimates of area, production and yield in respect of principal crops of foodgrains, oil seeds, sugar cane, fibres and important commercial and horticulture crops. These crops account for about 87 per cent of the total agricultural output. Estimates of area are based on a reporting system that is a mix of complete coverage and coverage by a sample, those of yield are based on a **system of crop cutting experiments** and **General Crop Estimation Surveys**. The preparation of these estimates takes time. Estimates of crop production are, however, needed earlier and in fact even

⁶ This part is taken from an article published in the economic times ET classrooms series, which can be accessed at http://articles.economictimes.indiatimes.com/2002-02-04/news/27340862_1_bananas-issue-prices-trade.

⁷ Report of the national Statistical Commission, Chapter 4.

before the crops are harvested, for policy purposes. Advance estimates of crop production are, therefore, made. The first assessment of the *kharif* crop is made in the middle of September, that is, when the South-West Monsoon is about to be over. The second advance estimate – a second assessment of the earlier estimate of the *kharif* crop and the first assessment of the *rabi* crop – is made in January. The third advance estimate is made at the end of March or the beginning of April and the fourth in June. The methodology for estimating area and yield of crops and for making advance estimates are given in **ASG**. Estimates of annual⁸ production of the principal crops mentioned above, gross area under different crops and yield per hectare of these crops and Index Numbers on these variables with base year the triennium ending 2007-08 = 100⁹ are also available in **ASG** in the form of a time series. So are estimates of production of crops by States.

Check Your Progress 1

- 1) Give two major sources of data on Agriculture and allied activities like dairy and fisheries.

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- 2) How can you retrieve the agricultural census data at district and tehsil level?

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- 3) How are annual estimates of crop production made by the Directorate of Statistics and Economics?

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⁸ Season-wise in some cases.

⁹ The base of these index numbers is a triennium in order to even out year to year to fluctuations of the underlying variable(s). The earlier base TE 1981-82 for these index number series was changed to TE 1993-94 in 2000-01 in order to update the index numbers and also for these to be in harmony with the index numbers of wholesale prices and industrial production with base 1993-94 = 100.

22.2.4 Livestock Census

The last quinquennial livestock census, the nineteenth in the series, was conducted in 2012. The census has collected information, district-wise on livestock, poultry, fishery and also agricultural implements. Livestock covers cattle, buffaloes, sheep, goats, pigs, horses and ponies, mules, donkeys, camels, yak, *mithun* and also, dogs and rabbits. These are classified by age (appropriate for the species), sex, breed, function ('breeding', 'work', 'both' and 'others' for males and 'in milk' or 'dry' in the case of females). Poultry covers cock, hen, duck, and drake, which are classified as *desi* and 'improved' varieties. Fishery covers fishing activity (inland capture, inland culture, marine capture and marine culture), members involved in fishing (male, female and child), persons engaged in fishing (part time male, full time male, part time female and full time female) craft/gear, namely, trawlers, gin-netters, liners, seiners, motorized canoe/*maran*, non-motorized canoe/*maran*, and miscellaneous gears like trawler-net, gill netter-net, cast-net, drag-net, hook lines, set barriers and others. Each of these classes except the last one is further classified into size (length) and horsepower. Agricultural implements cover annually operated implements, animal operated implements, plant protection equipment, water lifting devices, tractor and power operated implements, equipment for livestock and poultry and horticulture tools. Each group provides further data on four or five specific equipments like for example, thresher under the first group, ST Plough under the second, TPO Spray under the third, diesel under the fourth, Crawler Tractor under the fifth, incubator under the sixth and power operated tools under the seventh group.

The provisional results of the 2012 census are available on the website of the Department of Animal Husbandry, which can be accessed at <http://dahd.nic.in/dahd/WriteReadData/Livestock.pdf>. **ASG** and **BAHS** also present some data from the livestock census.

22.2.5 Data on Production of Major Livestock Products

AHSD is responsible for collection of statistics on animal husbandry, dairy and fisheries. These are published, as already mentioned, in **BAHS**. The latest relates to 2004. This presents data up to 2003-04, on

- i) long time series of estimates of production of milk, eggs, meat and wool;
- ii) State-wise short time series of such estimates;
- iii) State-wise and national estimates of per capita availability of milk and eggs;
- iv) contribution of cows, buffaloes and goats to milk production and of fowls and ducks to egg production in different States in 2003-04;
- v) average annual growth rates of production of major livestock products, milk, eggs and wool;
- vi) short time series on imports and exports of livestock and livestock products;
- vii) short time series on area under fodder crops, pastures and grazing in different States;
- viii) estimates of dry and green fodder production in different States during the three-year period ending 2002-03;

- ix) a short time series on the no. of artificial inseminations performed in different States;
- x) achievements in key components of dairy development;
- xi) livestock and poultry (numbers) in India and in different States – 1992, 1997 and 2003; and
- xii) time series on world livestock production and milk production.

22.2.6 Agricultural Statistics at a Glance (ASG)

The pocket book on agricultural statistics and the agricultural statistics at a glance are two annual publications available in the website http://eands.dacnet.nic.in/latest_2006.htm. One of the most important statistics, which is the statistics on Land utilization, is available in the **ASG**. As for inputs and access of individual operational holdings to such inputs, **ASG** provides data on:

- i) Methodology of crop estimation,
- ii) Socio-economic indicators: population and growth rates, State-wise classification of cultivators, Gross State Domestic Product for agriculture and allied activities,
- iii) Outlays, expenditure and capital formation in agriculture,
- iv) Area, production and yield of principal crops: target and achievement of production of major crops, three largest producing States of important crops, season-wise area, production and yield of food grains (rice, wheat, jowar, bajra, maize, total pulses, gram, tur, masur, nine oilseeds, groundnut, rapeseed and mustard, soyabean, sunflower), cotton, jute, mesta, sugarcane, tobacco, guarseed, etc.; yield rates of principal crops, area and yield under the high yielding varieties, area and production of horticulture and plantation crops (potato, onion, coconut, cashewnut, arecanut, garlic, ginger, sweet potato, turmeric, chillies, cardamom, pepper);
- v) All-India index numbers of area, production and yield of principal crops;
- vi) Area, production and yield of major crops in different countries;
- vii) Minimum support prices/marketed surplus ratios of various agricultural commodities;
- viii) Procurement by public agencies, per capita net availability of foodgrains, consumption and stocks of foodgrains and raw jute,
- ix) Import and export of agricultural commodities, tariffs and bound rates of major agricultural commodities, trends in wholesale price index of foodgrains and commercial crops;
- x) Land use statistics: agricultural land by use in India, selected categories of land use;
- xi) Inputs: production and use of agricultural inputs (fertilizers, crop-wise distribution of certified/quality seeds, consumption of electricity for agricultural purposes, flow of institutional credit for agricultural sector, state-wise number of kisan credit cards, state-wise/season-wise national agricultural insurance scheme, etc.

- xii) Agricultural census data: number and area of operational holdings by size group – marginal (size less than one HA¹⁰ area), small (1 to 2 HA), semi-medium (2 to 4 HA), medium (4 to 10 HA) and large (more than 10 HA), area irrigated by different source of irrigation and by size class and the position in different States;
- xiii) Estimated number of rural households, farmer households, indebted farmer households by size class of land possessed;
- xiv) Ceiling on land holding and wages for agricultural work;
- xv) Livestock population in India, all-India and State-wise production of milk, eggs, meat and wool;
- xvi) Fish production in India and States;
- xvii) Rainfall scenario and management of natural disaster.

The subjects **Subsidies** is a much-debated subject nowadays and agricultural subsidies in developing countries and developed countries of Europe and in USA are also much in the news. The Ministry of Agriculture provides a time series of the amount of subsidy given to agricultural with its break-up into subsidy for (i) fertilizers (ii) electricity, and (iii) irrigation¹¹, (iv) other subsidies given to marginal farmers and Farmers Cooperative Societies in the form of seeds, development of oil seeds, pulses, etc. One can easily derive the size of the total subsidy relative to GDP in current prices. The limitations to which these figures are subject need to be kept in mind.

22.2.7 Another Source of Data on Irrigation

The data on irrigation presented in **ASG** (referred to the proceeding sub-section) is based on crop statistics collected by **DESMOA** from village through a mix of a reporting system and sample surveys. The **Central Water Commission (CWC) under the Ministry of Water Resources** collects hydrological data on the important river systems in the country through 877 hydrological observation sites. The Ministry also conducts periodic **Censuses of Minor Irrigation Works** along with a sample check to correct the Census data. The **five Census** conducted so far related to **1986-87, 1993-94, 2000-01, 2006-07 and 2013-14**. This is conducted in all States/UTs except in Daman & Diu and Lakshadweep. The reports can be seen in the **website of the Ministry <http://micensus.gov.in>**. The **Report of the Minor Irrigation Census 2006-07** provides information on the 20.7 million minor irrigation works in the country like the share of ground water and surface water works, crop-wise utilization of the irrigation potential created, the manner in which the water is distributed in the field – sprinkler, drip, open channel or underground water irrigation and state-wise distribution of these variables.

The National Commission on Statistics had in their report made the following observations¹² on irrigation data available from **DESMOA** and the **Ministry of Water Resources**:

¹⁰ HA: abbreviation for hectare.

¹¹ The rates for supply of water to farmers are kept low as a matter of policy. This results in a loss to the Government Irrigation System. The excess of operating costs over gross revenue is treated as imputed irrigation subsidy.

¹² Chapter 4 of the Report.

- i) The **CWC** had a large volume of data on various aspects of irrigation without any statistical analysis of such data being carried out. The **CSO** and the **CWC** need to get together and ensure that the data collected is analysed and put to use by the statistical machinery.
- ii) There is a large variation between the statistics on the “area irrigated” published by **DESMOA** and the “irrigation potential actually utilized” published by the Ministry of Water Resources.
- iii) Data users should be made aware of the reasons for such variation – differences (if any) in concepts, definition etc.
- iv) The reluctance on the part of the State Governments to furnish data in view of their vested interest in the sharing of water is an added problem.
- v) Data from both the sources – **DESMOA** and the **Ministry of Water Resources** – are available after a large time lag and this needs to be reduced.

22.2.8 Other Data on the Agricultural Sector

Data on forest cover is, already mentioned, as part of land-use statistics presented on the basis of nine fold land-use classification in **ASG. Forest Survey of India (FSI)** also collects data on forest cover (dense forest, open forest and mangroves) through a biennial survey by making of Remote Sensing (RS) technology since 1987. The latest survey relates to 2013. Digital interpretation has reduced the time lag in the availability of such data obtained earlier through periodic reports from field formations. There are discrepancies between the data on forest area given by **ASG** and those given by **FSI** due to differences in the concepts and definitions used by the two organizations in collecting data. Data on production of industrial wood, minor forest produce and fuel wood are available with the **Principal Chief Conservator of Forests in the Ministry of Environment and Forests**.

The National Bank for Agriculture and Rural Development (NABARD) is another agency that is closely involved in agriculture, especially in the matter of credit to the sector through cooperatives. The **annual reports of NABARD and its publication like “Statistical Statements Relating to the Cooperative Movement in India”** (Part I deals with credit societies and the other with non-credit societies) and **“Key Statistics on Cooperative Banks”**, besides **NABARD’s website** (<https://www.nabard.org/english/home.aspx>) would be useful source of information on agricultural credit.

Statistics on area, production and yield of crops and subsidies given to agriculture are also available as time series in the **Economic Survey, expenditure budget documents of the Ministry of Finance and the RBI Handbook of Statistics on Indian Economy** (<http://rbidocs.rbi.org.in/rdocs/Publications/PDFs/000HSE13120914FL.pdf>). The third source is important in the context of the policy direction to banks and all financial institutions to ensure that 40 per cent of the total institutional credit should flow to the priority sector of which agricultural sector forms a part. The data warehouse of the RBI can be accessed at <http://dbie.rbi.org.in>. **The RBI Handbook of Statistics on Indian Economy – 2013-14** publishes time series data on:

- i) National income, saving and employment from 1952-53 onwards;

- ii) Output and prices of agricultural production, from 1956-57 onwards;
- iii) Assets and liabilities of the RBI, components of money stock, etc. from 1962-63 onwards;
- iv) Select aggregates of Scheduled Commercial Banks (SCBs), like demand deposit, time deposit, borrowings from RBI, liabilities to banks, investment in Government securities, food credit, non-food credit, cash in bank, balance with RBI, assets with banks from 1956-57 onwards;
- v) Sectoral deployment of non-food gross bank credit; industry-wise deployment of gross bank credit;
- vi) Direct and indirect institutional credit to agriculture and allied activities, 1975-76 onwards;
- vii) SCB's direct finance to farmers according to size of land holdings, 1985-86 onwards;
- viii) the distribution of non-food credit by priority sector credit and credit for other sectors;
- ix) the share of agriculture in priority sector credit;
- x) Financial market: structure of interest rate, etc.;
- xi) Public finances: key deficit indicators of central Government, major components of receipts of central Government, key deficit indicators of state governments, pattern of receipts of state governments, etc.;
- xii) Trade and balance of payment, etc.

Direct credit to agriculture is also available from other institutions and the Handbook furnishes data on:

- i) the short and long term loans issued by each of the institutions – Cooperatives, Scheduled Commercial Banks (SCBs), Regional Rural Banks (RRBs) and the State Governments directly to beneficiaries or borrows for agriculture and activities allied to it during a year; and
- ii) the loans outstanding at the end of each year.

Institutions like SCBs, Cooperative, RRBs and the Rural Electrification Corporation (REC) provide indirect credit to agriculture and allied activities. This is routed through some other agency, conduit or tier like the Electricity Boards (EBs), the Primary Agricultural Credit Societies (PACS), the Farmers' Service Societies (FSS), the Large-sized *Adivasi* Multi-Purpose Societies (LAMPS) and State-supported corporations and agencies. Such indirect credit is designed to promote agricultural productivity or to increase the agricultural income of the ultimate beneficiary, through a number of steps like distribution of fertilizers and other inputs to these activities, loans to EBs and loans to farmers through PACS, FSS and LAMPS. Other types of indirect credit include advances to State-supported corporations and agencies for onward lending of funds (up to Rs. 10,000/-) to weaker sections of society engaged in agriculture, namely, small and marginal farmers and those engaged in activities allied to agriculture.

We have already seen in the preceding Unit (on Macro-variable like national income) that **National Accounts Statistics (NAS)** publishes data on the contribution of agriculture and its sub-sectors to GDP and other measures of